COMPUTER NETWORKS - UE24CS252B

Demonstration Of Basic Commands

1. ipconfig

- **Syntax**:

```
ipconfig [options]
```

For Mac/Linux:

```
ifconfig [options]
```

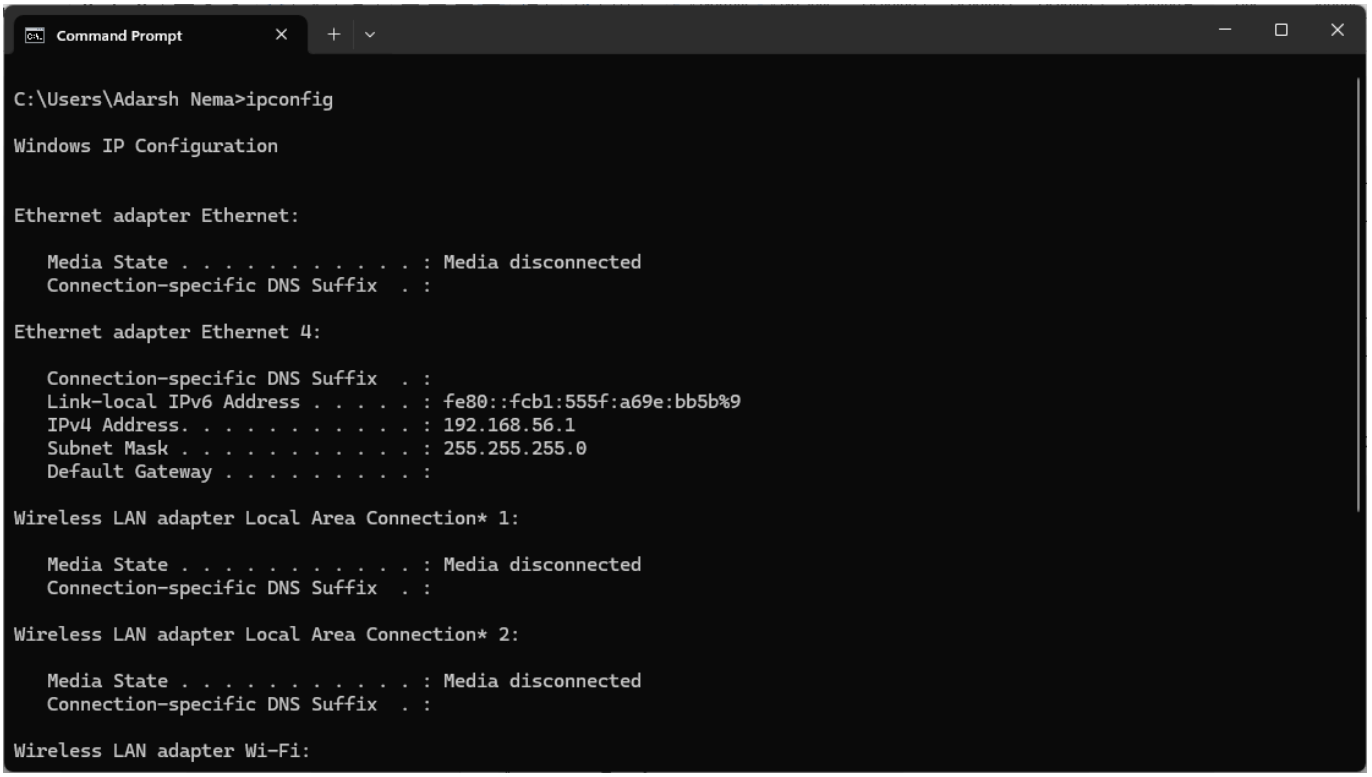
Example:

```
ipconfig /all
```

For Mac/Linux:

```
ifconfig -a
```

- **What It's Used For**:
  - Displays network configuration details like IP address, subnet mask, and default gateway.
  - Can be used to release or renew IP addresses (/release, /renew).
- **Why It's Important**:
  - Helps diagnose and resolve network configuration issues, such as incorrect IP settings or connectivity problems.



For Mac/Linux:

```
zap@yabsolutely $>ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
        options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
        inet 127.0.0.1 netmask 0xff000000
        inet6 ::1 prefixlen 128
        inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
        nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 56:71:9a:df:3d:d3
        media: none
        status: inactive
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 56:71:9a:df:3d:d4
        media: none
        status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 56:71:9a:df:3d:b3
        nd6 options=201<PERFORMNUD,DAD>
```

2. ping (Same for Mac/Linux)

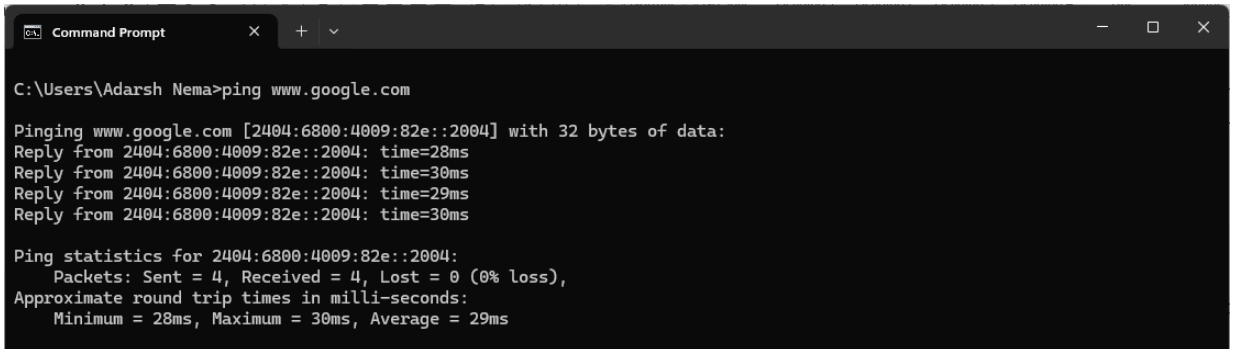
- **Syntax**:

```
ping [hostname or IP address]
```

Example:

```
ping google.com
```

- **What It's Used For**:
  - Tests connectivity between your computer and a remote server.
  - Measures packet loss and latency.
- **Why It's Important**:
  - Quickly identifies if a host is reachable and helps troubleshoot connection problems.



For Mac/Linux:

```
[zap@yabsolutely $>ping google.com
PING google.com (142.251.43.142): 56 data bytes
64 bytes from 142.251.43.142: icmp_seq=0 ttl=118 time=16.731 ms
64 bytes from 142.251.43.142: icmp_seq=1 ttl=118 time=19.552 ms
64 bytes from 142.251.43.142: icmp_seq=2 ttl=118 time=24.124 ms
64 bytes from 142.251.43.142: icmp_seq=3 ttl=118 time=16.102 ms
64 bytes from 142.251.43.142: icmp_seq=4 ttl=118 time=19.435 ms
64 bytes from 142.251.43.142: icmp_seq=5 ttl=118 time=19.596 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 16.102/19.257/24.124/2.590 ms
```

3. netstat (Same for Mac/Linux)

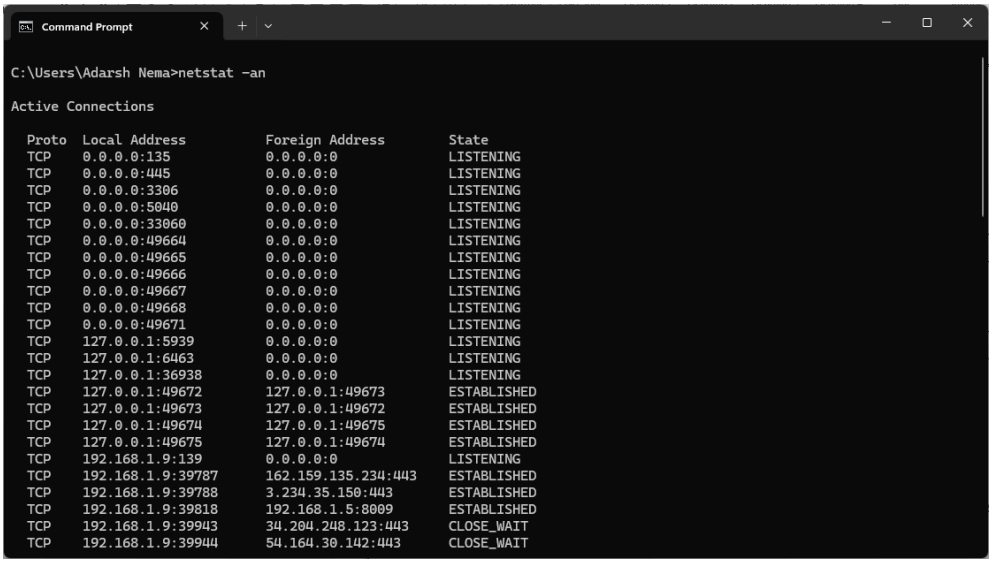
- **Syntax**:

```
netstat [options]
```

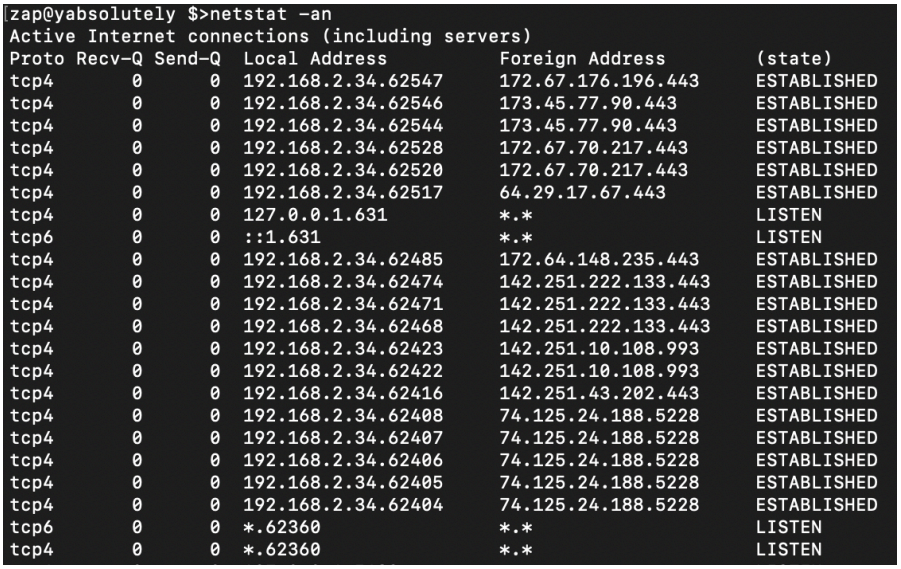
Example:

```
netstat -an
```

- **What It's Used For**:
  - Displays active network connections, listening ports, and protocol statistics.
  - Can show processes using specific ports with -b option.
- **Why It's Important**:
  - Useful for monitoring network activity and identifying applications consuming bandwidth or exposing vulnerabilities.



For Mac/Linux:



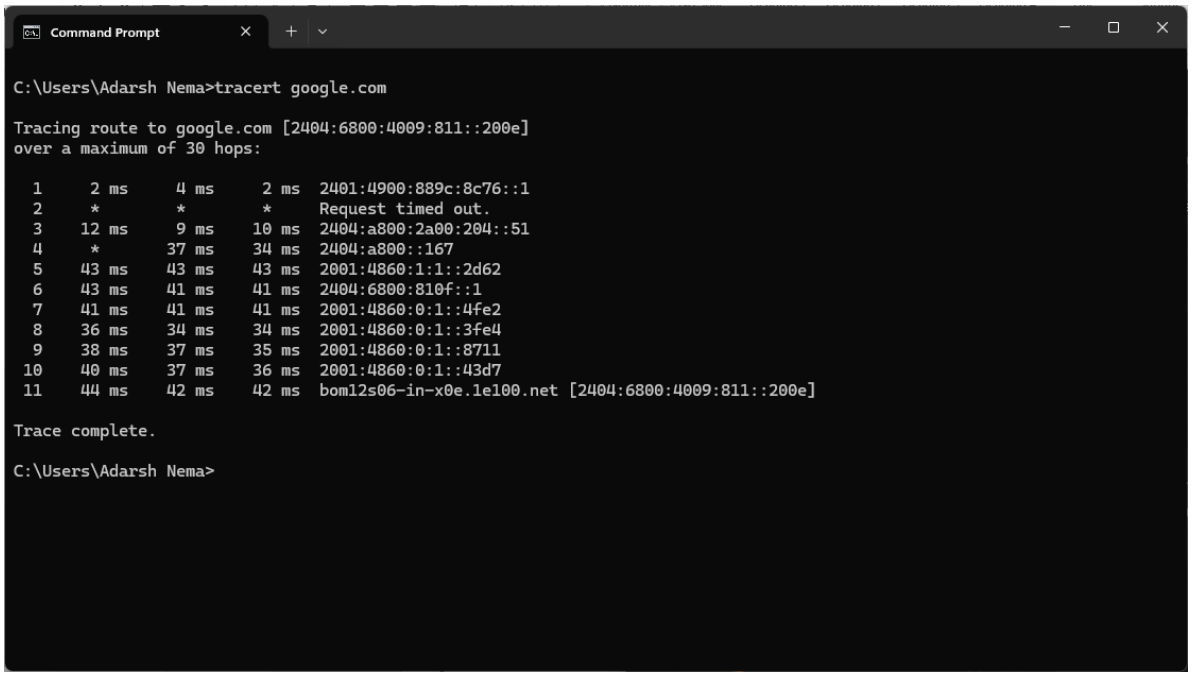
4. tracert/traceroute

- **Syntax**:

```
tracert [hostname or IP address]
```

Example:

```
tracert google.com
```



For Mac/Linux:

```
traceroute [hostname or IP address]
```

```
traceroute  HYPERLINK "http://google.com"google.com
```

```
zap@yabsolutely $>traceroute google.com
traceroute to google.com (142.251.222.206), 64 hops max, 40 byte packets
 1  192.168.2.1 (192.168.2.1)  6.693 ms  4.970 ms  2.521 ms
 2  dsldevice.lan (192.168.1.1)  7.318 ms  3.346 ms  3.431 ms
 3  abts-kk-dynamic-1.72.76.171.airtelbroadband.in (171.76.72.1)  8.780 ms  8.425 ms  8.975 ms
 4  125.21.0.185 (125.21.0.185)  8.101 ms
    125.21.0.149 (125.21.0.149)  8.163 ms
    nsg-corporate-193.159.186.122.airtel.in (122.186.159.193)  7.591 ms
 5  116.119.68.108 (116.119.68.108)  14.739 ms *
    116.119.57.99 (116.119.57.99)  37.439 ms
 6  * * *
 7  * * *
 8  142.251.55.206 (142.251.55.206)  19.883 ms * *
 9  142.251.51.118 (142.251.51.118)  18.750 ms
    142.250.208.230 (142.250.208.230)  18.028 ms
    142.251.51.118 (142.251.51.118)  15.143 ms
10  172.253.70.167 (172.253.70.167)  16.717 ms
    142.251.55.240 (142.251.55.240)  13.273 ms
    216.239.56.70 (216.239.56.70)  15.470 ms
11  142.251.55.239 (142.251.55.239)  14.973 ms
    hkg07s55-in-f14.1e100.net (142.251.222.206)  21.092 ms
    142.250.63.172 (142.250.63.172)  13.788 ms
```

5. nslookup

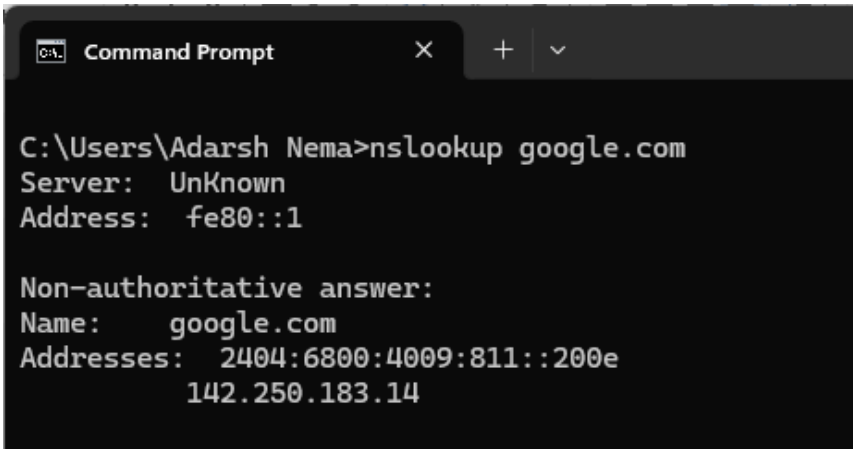
- **Syntax**:

```
nslookup [hostname or IP address]
```

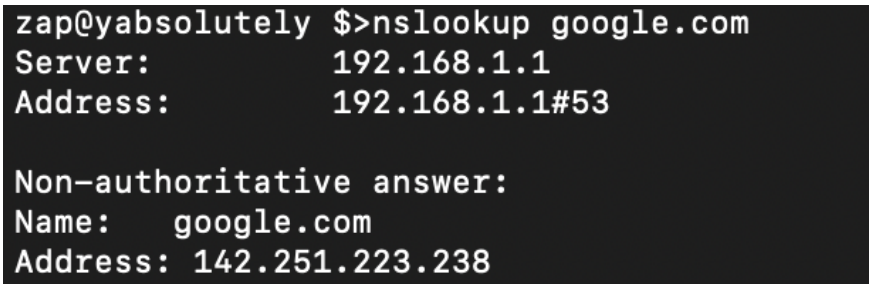
Example:

```
nslookup example.com
```

- **What It's Used For**:
  - Queries the DNS to resolve domain names into IP addresses and vice versa.
  - Provides DNS record details.
- **Why It's Important**:
  - Diagnoses DNS-related issues like domain resolution failures or DNS server unavailability.



For Mac/Linux:



6. arp

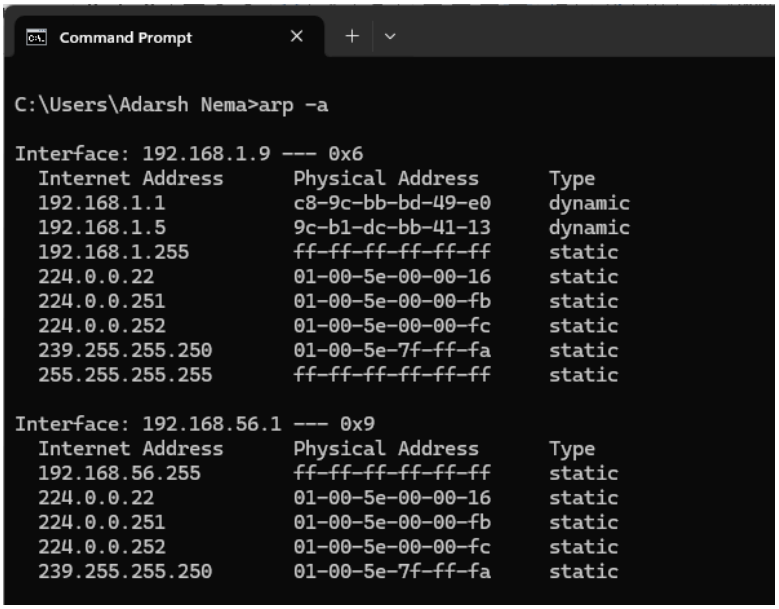
- **Syntax**:

```
arp [options]
```

Example:

```
arp -a
```

- **What It's Used For**:
  - Displays or modifies the ARP (Address Resolution Protocol) cache.
  - Shows mappings between IP addresses and MAC addresses.
- **Why It's Important**:
  - Helps troubleshoot issues with device communication on a local network.



For Mac/Linux:

```
[zap@yabsolutely $>arp -a
? (192.168.2.1) at ec:22:80:d0:f:74 on en0 ifscope [ethernet]
? (192.168.2.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
```

7. netsh

- **Syntax**:

```
netsh [context]
```

Example:

```
netsh wlan show profiles
```

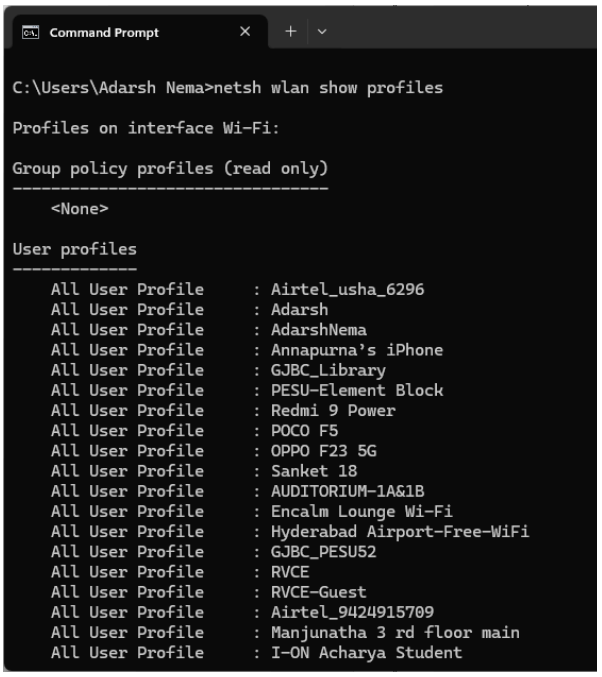
For Mac:

```
networksetup [options]
```

For Linux:

```
nmcli [options]
```

- **What It's Used For**:
  - Configures and monitors Windows networking settings.
  - Useful for managing wireless profiles, firewall rules, and network adapters.
- **Why It's Important**:
  - Enables advanced network troubleshooting and configuration directly from the command line.



For Mac:

```
zap@yabsolutely $>networksetup -listallnetworkservices
An asterisk (*) denotes that a network service is disabled.
USB 10/100/1000 LAN
Thunderbolt Bridge
Wi-Fi
zap@yabsolutely $>networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 192.168.2.34
Subnet mask: 255.255.255.0
Router: 192.168.2.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: 2c:ca:16:4b:72:5e
```

8. getmac

- **Syntax**:

```
getmac [options]
```

Example:

```
getmac /v
```

For Mac/Linux:

```
ifconfig [network name] | grep ether
```

Example

```
ifconfig en0 | grep ether
```

```
[zap@yabsolutely $>ifconfig en0 | grep ether
        ether 3a:a7:41:1e:c0:ee
```

- **What It's Used For**:
  - Displays the MAC addresses of network interfaces.
  - Shows details of enabled network adapters.
- **Why It's Important**:
  - Essential for verifying the hardware addresses of network cards, especially when troubleshooting connectivity or security issues.

9. tasklist

- **Syntax**:

```
tasklist [options]
```

Example:

```
tasklist /svc
```

For Mac/Linux:

```
ps aux
```

- **What It's Used For**:
  - Lists all running processes on the system, including those associated with networking tasks.
- **Why It's Important**:
  - Identifies processes using high network resources or conflicting with other applications.

10. route

- **Syntax**:

```
route [options]
```

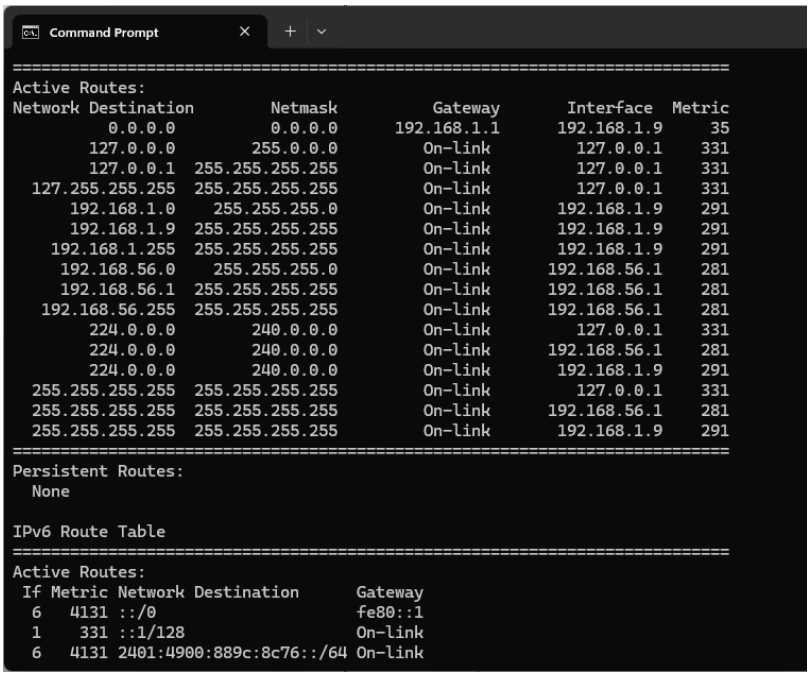
Example:

```
route print
```

For Mac/Linux:

```
netstat -rn
```

- **What It's Used For**:
  - Displays and modifies the routing table.
  - Can be used to add or delete static routes.
- **Why It's Important**:
  - Helps troubleshoot routing issues in complex network setups, such as multiple gateways or VPN connections.



For Mac/Linux:

```
zap@yabsolutely $>netstat -rn
Routing tables

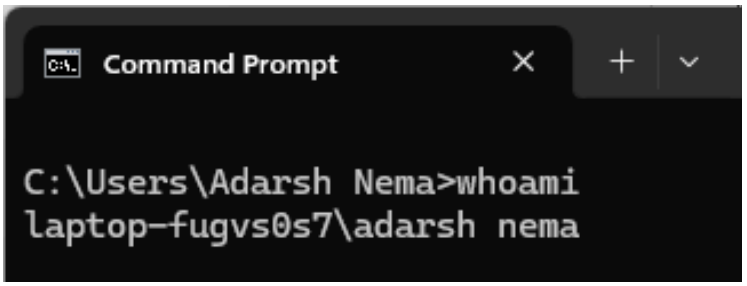
Internet:
Destination        Gateway            Flags               Netif Expire
default            192.168.2.1        UGScg                 en0
127                127.0.0.1          UCS                   lo0
127.0.0.1          127.0.0.1          UH                    lo0
169.254            link#11            UCS                   en0      !
192.168.2          link#11            UCS                   en0      !
192.168.2.1/32     link#11            UCS                   en0      !
192.168.2.1        ec:22:80:d0:f:74   UHLWIir               en0   1176
192.168.2.34/32    link#11            UCS                   en0      !
192.168.2.255      ff:ff:ff:ff:ff:ff  UHLWbI                en0      !
224.0.0/4          link#11            UmCS                  en0      !
224.0.0.251        1:0:5e:0:0:fb      UHmLWI                en0
255.255.255.255/32 link#11            UCS                   en0      !
```

11. whoami (Same For Mac/Linux)

- **Syntax**:

```
whoami
```

- **What It's Used For**:
  - Displays the username and domain of the currently logged-in user.
- **Why It's Important**:
  - Helps verify identity and permissions, especially when troubleshooting user- specific network or access issues.



12. powercfg/pmset -g/upower -d

- **Syntax**:

```
powercfg [options]
```

Example:

```
powercfg /energy
```

For Mac:

```
pmset -g
```

For Linux:

```
upower -d
```

- **What It's Used For**:
  - Analyzes power consumption and provides a report on energy efficiency.
- **Why It's Important**:
  - Useful for diagnosing power issues that may affect network device performance, such as power-saving modes disabling network adapters.

13. systeminfo

- **Syntax**:

```
systeminfo
```

For Mac/Linux:

```
uname -a
```

- **What It's Used For**:
  - Displays detailed information about the system, including network configuration and updates.
- **Why It's Important**:
  - Useful for gathering system details during network troubleshooting or hardware compatibility checks.

14. sc

- **Syntax**:

```
sc [command] [service_name]
```

Example:

```
sc query dnscache
```

For Mac:

```
launchctl [options]
```

For Linux:

```
systemctl [options]
```

- **What It's Used For**:
  - Manages Windows services, such as starting, stopping, or querying the status of networking-related services.
- **Why It's Important**:
  - Critical for diagnosing issues with services like DNS, DHCP, or networking drivers.

15. telnet (Same for Mac/Linux)

- **Syntax**:

```
telnet [hostname or IP address] [port]
```

Example:

```
telnet example.com 80
```

- **What It's Used For**:
  - Connects to remote servers using the Telnet protocol to test open ports or communicate with services.
- **Why It's Important**:
  - Useful for diagnosing server communication or port accessibility issues.

16. fsutil/diskutil

- **Syntax**:

```
fsutil behavior set disable8dot3 1
```

For Mac:

```
diskutil list
```

- **What It's Used For**:
  - Performs advanced file and volume management tasks, such as managing filesystems that might interact with network shares.
- **Why It's Important**:
  - Useful for ensuring optimal performance when using shared network drives or distributed filesystems.

17. wmic

- **Syntax**:

```
wmic [context]
```

Example:

```
wmic nic get name, macaddress
```

- **What It's Used For**:

- Executes Windows Management Instrumentation (WMI) queries for detailed information about network adapters and other hardware.

- **Why It's Important**:
  - Provides low-level details about hardware and network interfaces that can assist in troubleshooting.

NOTE: there is no single command for wmic in macOS. To know about the hardware and others you can use networksetup.

18. net user

- **Syntax**:

```
net user [username]
```

Example:

```
net user Administrator
```

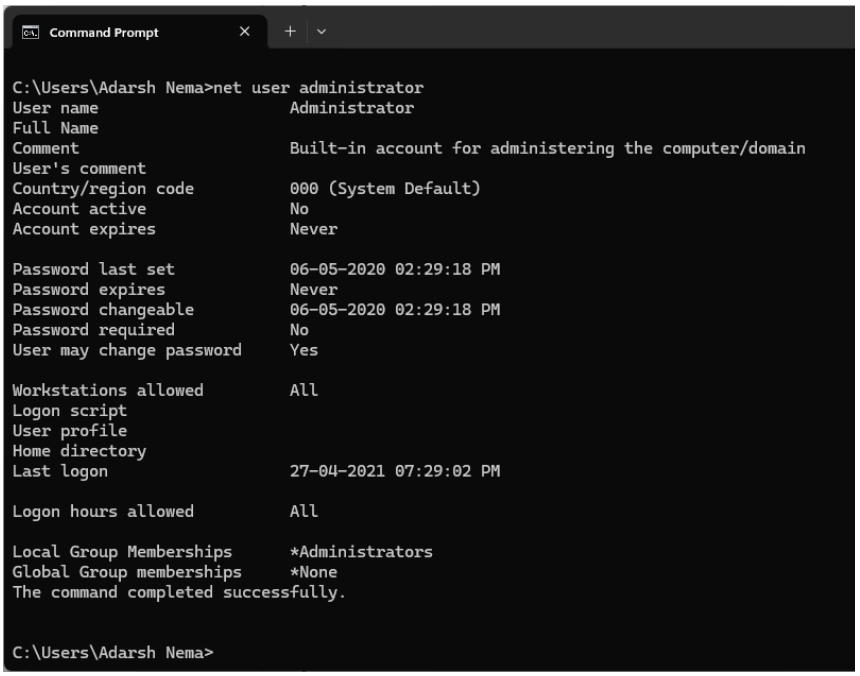
For Mac:

```
dscl . -read /Users/<username>
```

For Linux:

```
lslogins -u <username>
```

- **What It's Used For**:
  - Displays or manages user account details, such as permissions and login status.
- **Why It's Important**:
  - Helps verify or configure user accounts that might impact network resource access or privileges.


Command Prompt
C:\Users\Adarsh Nema>net user administrator
User name Administrator
Full Name
Comment Built-in account for administering the computer/domain
User's comment
Country/region code 000 (System Default)
Account active No
Account expires Never
Password last set 06-05-2020 02:29:18 PM
Password expires Never
Password changeable 06-05-2020 02:29:18 PM
Password required No
User may change password Yes
Workstations allowed All
Logon script
User profile
Home directory
Last logon 27-04-2021 07:29:02 PM
Logon hours allowed All
Local Group Memberships *Administrators
Global Group memberships *None
The command completed successfully.
C:\Users\Adarsh Nema>